Adaptive trace estimation under a limited computation budget

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Project Overview

My UROP project investigated adaptive query allocation in Hutch++, a randomized algorithm for estimating the trace of a matrix using only matrix-vector products. The trace is obtained by adding the diagonal entries of a matrix, which is easy when those entries are directly available. However, computing the trace can be expensive when the matrix is large and is available only through its action on a vector. In this setting, each matrix-vector multiplication is called a query. The challenge is to estimate the trace accurately using a limited number of queries.

Hutch++ estimates the trace in two stages. First, it uses some queries to construct a low-dimensional subspace that captures important matrix directions. It then projects the matrix onto this subspace and uses random probes to estimate the trace of the remaining part [1, 2]. My project builds on this method by studying whether the query budget can be allocated adaptively between these stages. I began exploring the project around June 2026, hoping to find a faster or more accurate version of the method. My central question became whether an initial set of observations could help choose a better division of queries between these stages. As the investigation developed, I also asked whether evaluating a proposed adjustment was worth the queries needed to make that decision.

The project combined mathematical analysis with controlled numerical experiments. Together, these analyses showed that an adaptive method may improve one measure of performance while becoming worse under another, especially when the queries used for evaluation count toward the total budget. Capturing more structure can reduce one source of error while leaving fewer samples for the final estimate. Moreover, determining whether an adjustment actually helps consumes queries from the same budget, which can further reduce the number of samples available for the final estimate.

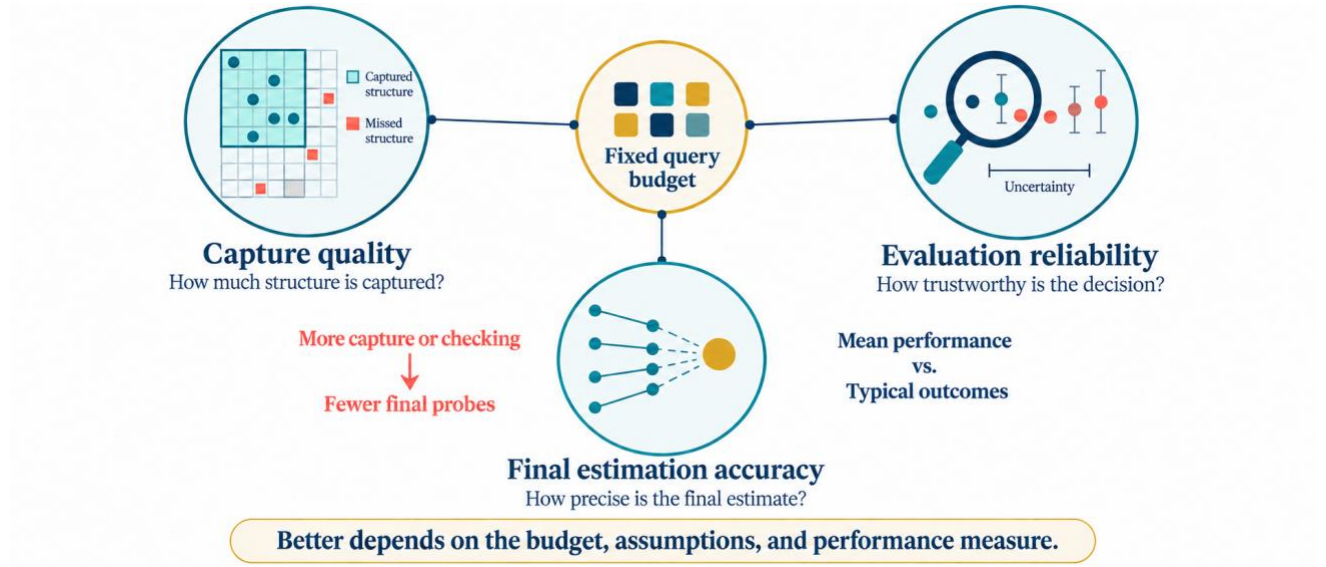


Figure 1. The central trade-off in adaptive trace estimation. Capture quality, evaluation reliability, and final estimation accuracy draw on a shared query budget. An improvement in one aspect can leave fewer resources for another. Conceptual illustration; no experimental values are shown.

Assessment of Objectives

I developed and evaluated adaptive allocation ideas using matrices with controlled spectral structure, while tracking matrix-vector queries to make comparisons meaningful.

I derived a condition for when adding one more direction to the subspace is worthwhile. However, the added direction must reduce the residual variance enough to make up for the two queries taken from the final estimate. I also distinguished between an ideal model, which assumes that the leading eigenvectors are captured, and the actual error produced by a randomized sketch. A decreasing spectral tail alone does not show that the modified method will produce a more accurate trace estimate. The confidence analysis used existing mathematical results to control the variability of randomized trace estimates, including bounds for random quadratic forms [3] and hypercontractive moment estimates for random-sign polynomials [4].

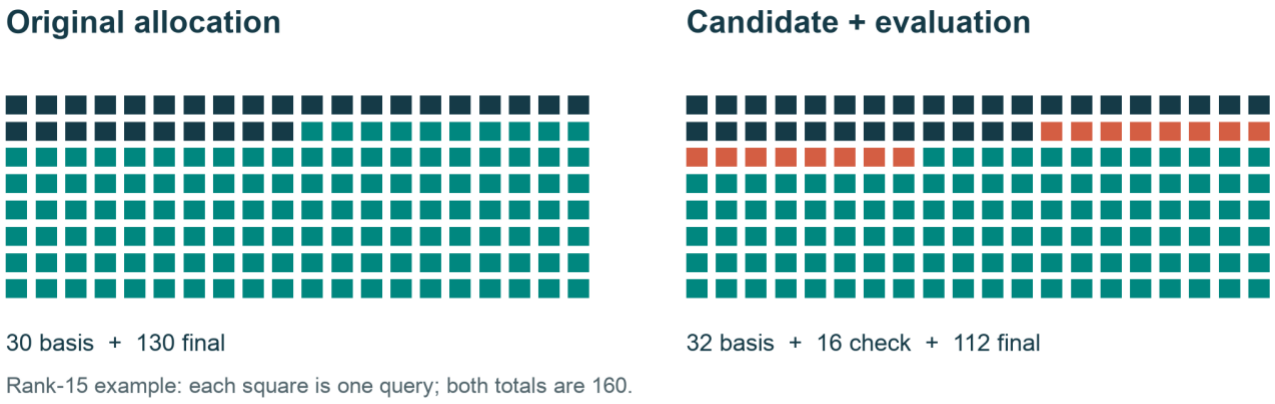


Figure 2. A 160-query example at baseline rank 15: extending the basis and checking the candidate leaves 112 final probes rather than 130. Teal: final probes; navy: basis construction and trace computation; coral: evaluation.

Experiments and a separate algebraic example documented in the project records [5] showed why distinguishing ideal subspace capture from the behavior of a randomized sketch matters. A sketch can have the expected numerical rank and display a pronounced spectral gap while missing an important direction. I therefore investigated direct estimates of residual risk, meaning expected squared estimation error after the basis is fixed. In one controlled study, a decision rule based on sample variance correctly identified approximately 79.7% of the eligible high-risk cases in which the candidate actually performed better. These cases were selected because the original baseline was among the 5% highest-risk cases within its comparison group and the candidate had lower true risk. This percentage describes a selected group of cases, rather than an overall improvement rate or a mathematical guarantee.

The most revealing result appeared when evaluation queries were charged to the total budget. In that study, 575 of 600 fixed randomized basis paths [5] had higher expected squared error after paying for the procedure than their original baseline. However, the average risk across all paths improved substantially because a small number of severe baseline failures were reduced. Each candidate was compared with its corresponding baseline allocation, so this result does not establish superiority over every standard Hutch++ allocation. These observations showed why average performance and the experience of a typical run can lead to different conclusions.

The project clarified a central trade-off in adaptive allocation: capturing more structure can reduce residual error, but constructing or evaluating that improvement uses queries that could otherwise support the final estimate. I derived mathematical conditions for this trade-off, implemented the corresponding allocation procedures, and evaluated them through controlled experiments. The original proposal also included designing structured sketch projections. The completed work focused on query allocation, how reliably randomized sketches capture important directions, and how the cost of evaluating a candidate affects the final estimate. Structured sketch design remains a possible continuation.

Student Reflection

The hardest part was formulating a concrete research problem. The proofs were challenging as well, but they became more manageable once the problem and its assumptions were clear. At that point, many of the lemmas needed for the analysis were standard. An early argument suggested that reducing the spectral tail should improve performance without examining the full error expression. I found that this conclusion was incomplete by working through the full expression and testing it with counterexamples. I discussed some findings with my mentor. This helped me recognize the competing effects of reducing residual variance and reducing the number of samples available to estimate it.

That experience changed how I approach mathematical research. I now examine theoretical assumptions one by one and ask whether I can justify every step of an argument. When results disagree, I consider counterexamples, different random initializations, and possible problems with the experimental design. If those checks do not explain the discrepancy, I repeat the numerical checks when possible before reconsidering the research question itself. A proposed change to an argument may suggest a next step, but I still need to verify it mathematically or experimentally. I also learned that I need to understand the purpose of each part of an implementation and how the parts interact before I can assess or debug its results.

The project made the meaning of "better" more precise for me. A method may improve average error while making most runs slightly worse, or become more reliable only under additional assumptions. I found this interaction between constraints, uncertainty, and competing objectives especially interesting. Developing mathematical results and catching flawed arguments were both meaningful accomplishments. The experience strengthened my interest in randomized numerical methods and optimization, where careful reasoning can establish useful conclusions despite uncertainty.

Closing Remarks

Future work could quantify the trade-off more precisely, tighten the confidence bounds, or improve how the initial sketch captures structure. One possibility is to investigate leverage-guided sampling, which uses information about a subspace to guide which coordinates or columns are sampled. Related leverage-score methods exist for low-rank approximation [6]. Their value for this trace-estimation setting remains an open question: estimating the scores costs resources, and scores derived from an incomplete sketch may overlook missing directions. I would evaluate any such proposal under the same total query budget. I am grateful for Professor Padmanabhan's guidance and the opportunity to develop greater independence in research.

References

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